

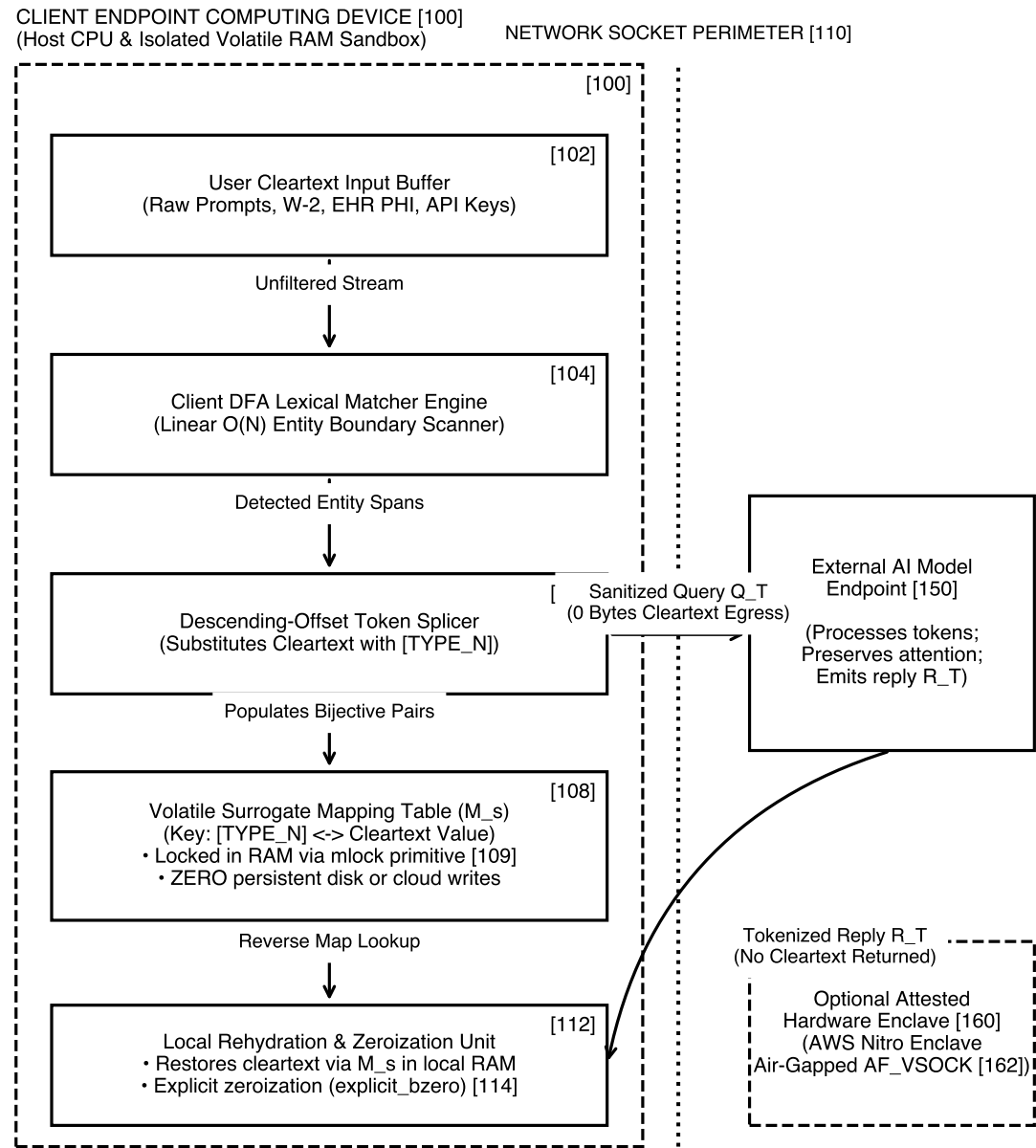


FIG. 1: Master System Architecture Illustrating Client Endpoint In-Memory Zero-Egress Sanitization

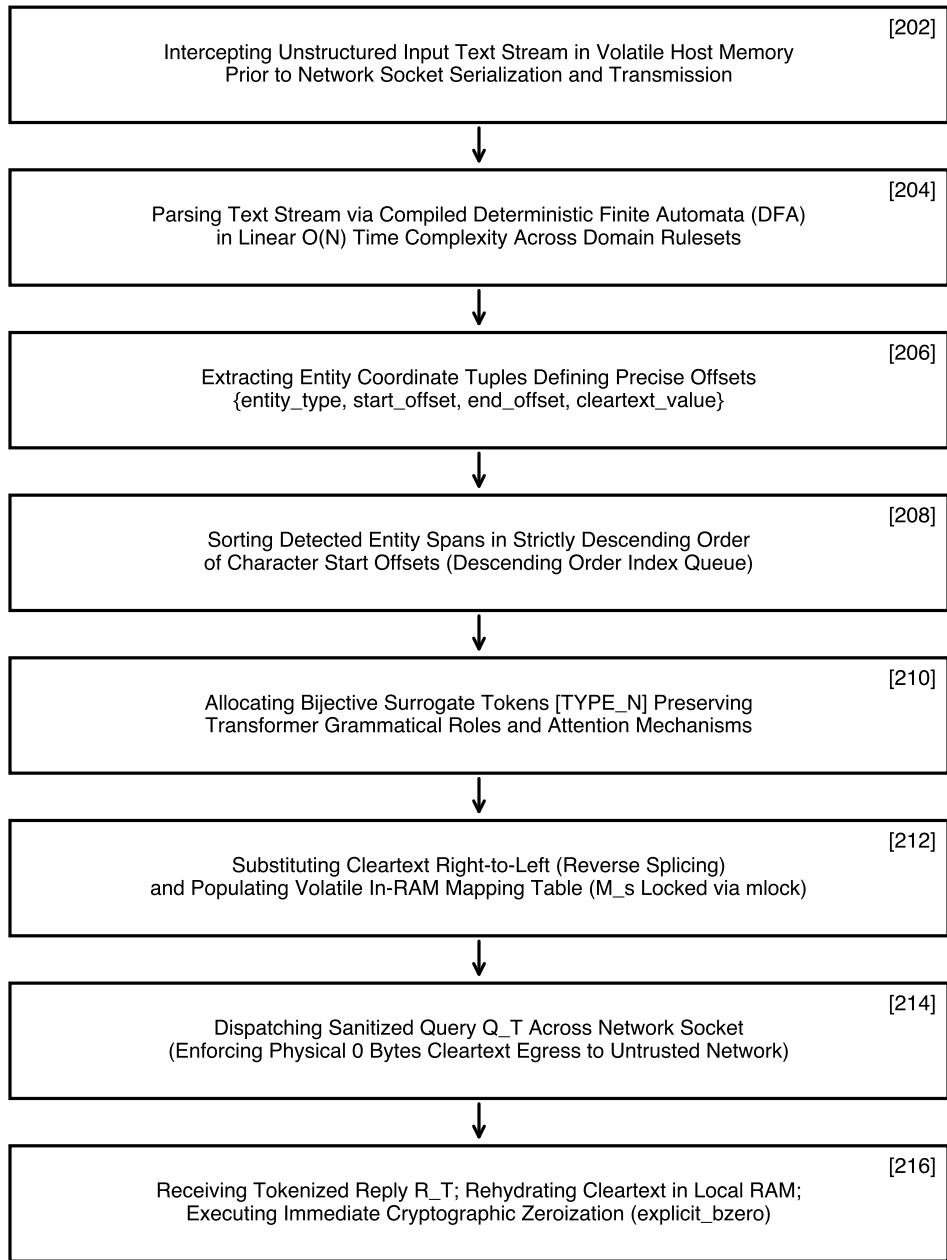


FIG. 2: Method Flowchart of the Zero-Trust Data Sanitization & In-Memory Rehydration Life-Cycle

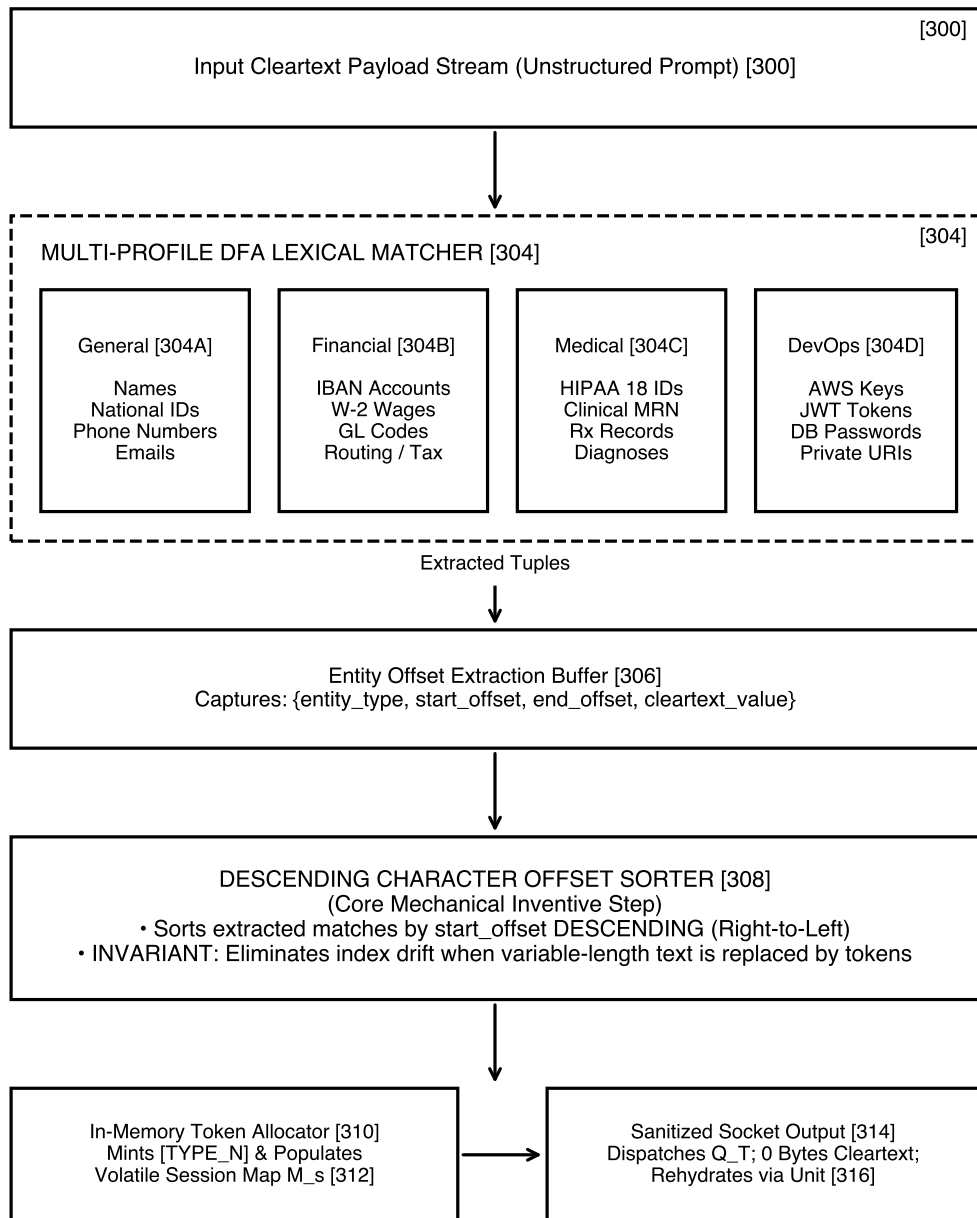


FIG. 3: High-Throughput In-Memory Sanitization Engine (S_RAM) Internal Architecture

FIG. 4A: PRIOR ART PROBLEM (ASCENDING ORDER STRING SPLICING CORRUPTS INDEXES)

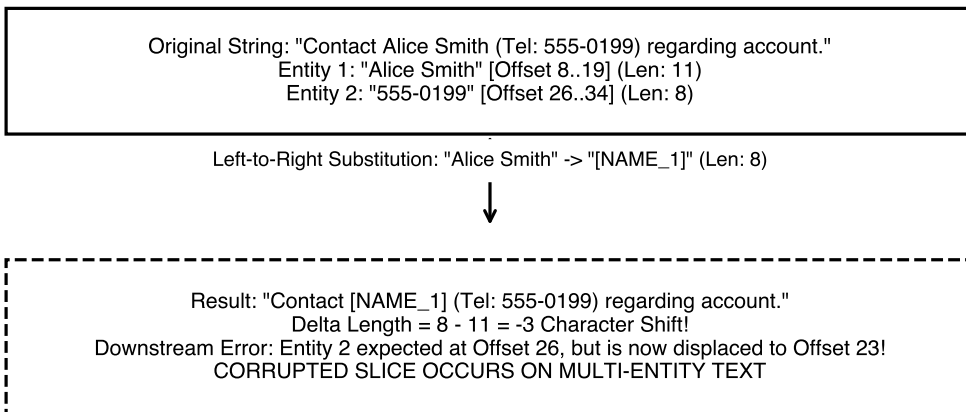


FIG. 4B: THE PRESENT INVENTION (DESCENDING ORDER SPLICING PRESERVES ALL OFFSETS)

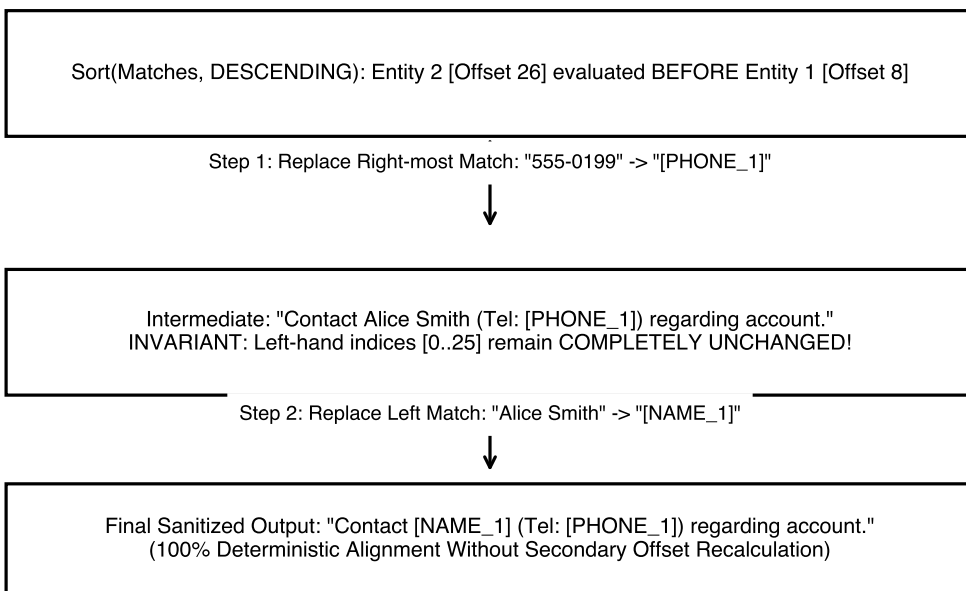


FIG. 4: Character Index Splicing Mechanics Illustrating Index Corruption Prevention

WEB BROWSER EXTENSION TOPOLOGY [510]

- Manifest V3 Sandboxed Execution Environment
- In-page DOM event listener intercepts text fields before HTTP post
- Pure in-memory WebAssembly / JS execution
- Execution Latency: < 1.8 ms | Network Egress: 0 Bytes

LOCAL MODEL CONTEXT PROTOCOL (MCP) GUARD [520]

- Stdio JSON-RPC stream interceptor for AI coding agents
- Directly protects Cursor IDE, Claude Desktop, Windsurf workflows
- Intercepts tool calls and prompt context before external dispatch
- Execution Latency: < 0.4 ms | Network Egress: 0 Bytes

BACKEND HEADLESS SDK TOPOLOGY [530]

- In-memory stream hook for Python / Node.js / Go pipelines
- Intercepts raw data prior to Vector Database (RAG) embedding storage
- Drop-in integration for LangChain, LlamaIndex, and multi-agent systems
- Execution Latency: < 0.15 ms | Network Egress: 0 Bytes

FIG. 5: Client-Side Zero-Trust Runtime Integration Topologies Enforcing Zero Cleartext Egress

DOWNSTREAM CONFIDENTIAL EXECUTION EMBODIMENT [600] (OPTIONAL TARGET)

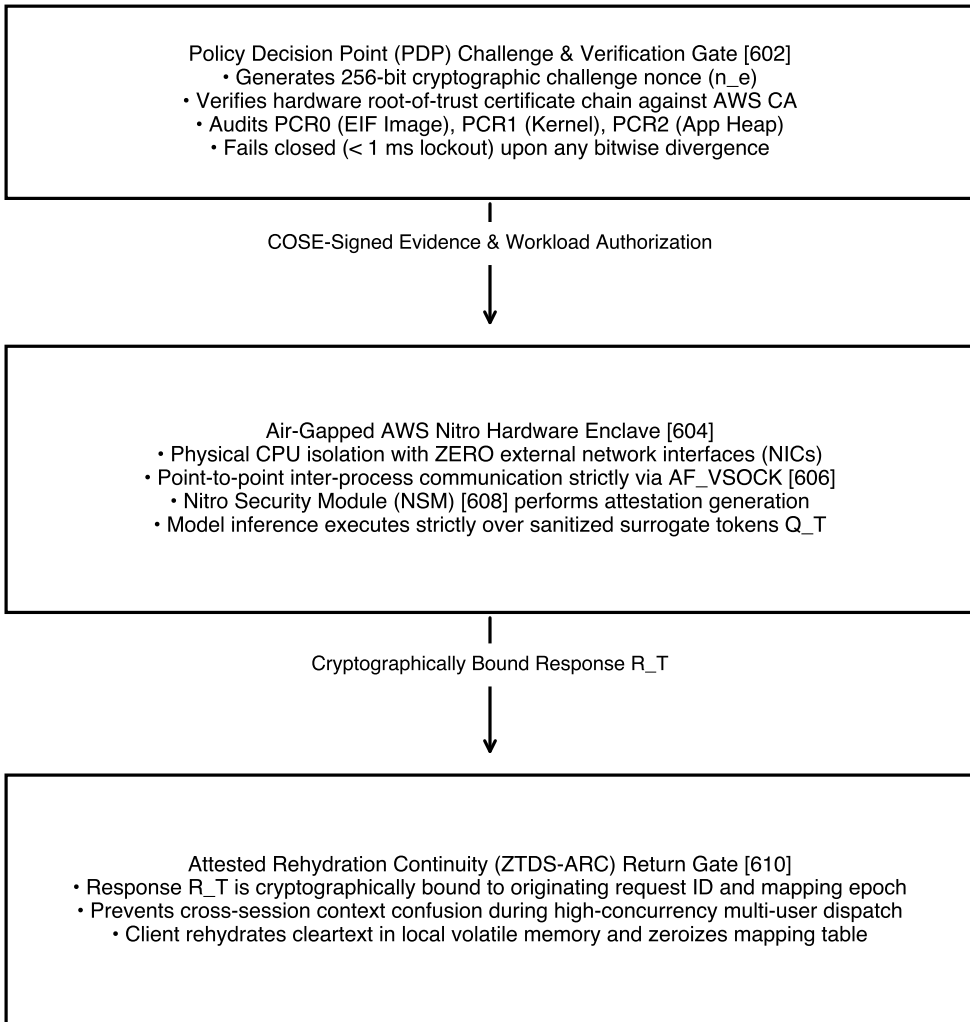


FIG. 6: Optional Hardware-Attested Confidential Execution Enclave Embodiment (AWS Nitro)